Counting and Binomial Theorem

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In these notes, we will explore the principles and applications of the Binomial Theorem, which can be a tremendously powerful tool for breaking down complex problems into manageable tasks.

Combinations and Permutations

First, we will discuss a few principles at the heart of combinatorics, the study of counting and probability. Consider 4 people standing in a line at Starbucks. The number of ways to order the people in the line is $4! = 4 * 3 * 2 * 1 = 24$, since there are 4 people to select for the first slot, 3 people for the next one, 2 people for the one after that, and finally 1 person for the last slot.

Figure 1: People lining up for Starbucks diagram.

Now, suppose that only two out of these four people are standing in line at a time. The number of distinct arrangements of people is $4 * 3 = 12$. We note that $12 = \frac{4!}{2!}$, and in this context, we have 4 total people and we are arranging $4 - 2 = 2$ of them at a time. In fact, if we consider the case where only one of these four people is there at a time, there are $4 = \frac{4!}{3!} = \frac{4!}{(4-1)!}$ arrangements.

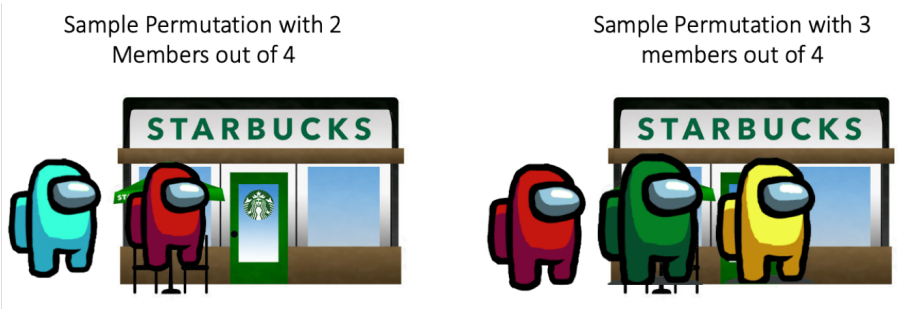


Figure 2: Illustration of the permutation concept.

We call an ordered arrangement of k members from a set of n total elements a *permutation*, whose formal definition is expounded on below.

Permutation

A permutation of a set is the ordering of its members into a sequence. The number of ways to select k elements from a n-element set and arrange them into a sequence can be expressed as $P(n, k)$, where

$$P(n, k) = n * (n - 1) * ... * (n - k + 1) = \frac{n!}{(n - k)!}.$$

There are numerous applications related to permutations. Operationalizing this concept, we will solve a few problems from the AMC 10.

Example 1

From a collection of art pieces, there are 504 ways to select and arrange three pieces in a row on a particular wall. if the next wall may hold the remaining pieces in a row, how many ways may the remaining pieces be arranged? (MN Math League 2016/17)

Solution: Let n be the number of total art pieces. The number of ways to select and arrange three pieces is a permutation of 3 elements from a set of n total, or $P(n, 3)$. Thus, $P(n, 3) = 504$ so $\frac{n!}{(n-3)!} = 504$. We can expand the left hand side as shown.

$$\frac{n!}{(n-3)!} = \frac{n * (n-1) * (n-2) * (n-3)!}{(n-3)!} = n(n-1)(n-2).$$

As $n(n-1)(n-2) = 504$ and n is an integer, we quickly experiment with a few integers to find that $n = 9$. There are $9 - 3 = 6$ remaining pieces, which can be arranged in a total of $6! = \boxed{720}$ ways.

Example 2

How many ways can a student schedule 3 mathematics courses – algebra, geometry, and number theory – in a 6-period day if no two mathematics courses can be taken in consecutive periods? (What courses the student takes during the other 3 periods is of no concern here.) (AMC 10A 2018)

Solution: We determine which of the three periods will be occupied by mathematics course. From there, we can count the number of arrangements of the three math classes across these periods. First, we find that there are four ways that the periods can be selected so that there are no consecutive class periods, as illustrated below.

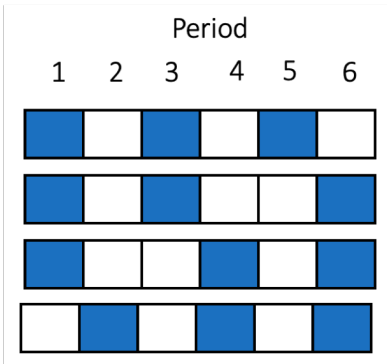


Figure 3: The 4 possibilities for selecting the three slots to occupy with classes.

For each of these four cases, the 3 classes can be sorted into the slots in $3! = 3 * 2 * 1 = 6$ ways. Thus, the total number of ways to schedule the classes is $4 * 3! = \boxed{24}$.

We have primarily discussed selecting k elements out of n and then *arranging them into an ordered sequence*. The next task is to count the number of ways to select k elements, disregarding the order in which that they are selected or arranged sequentially. A selection of k elements out of n, neglecting order, is called a combination.

Combination

The number of ways to select k elements from a n-element set can be expressed as $C(n,k) = \binom{n}{k}$, where

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}.$$

Note that $\frac{P(n,k}{C(n,k)} = k!$, which means that given the same n and k, there are $k!$ more permutation than combinations. This is because combinations discount—no pun intended—the arrangement of the k items after selecting them, which can transpire in $k!$ ways. The principle of combinations is a quite useful mathematical tool and we can furnish our toolbox with some additional properties.

Combination Properties

$$\binom{n}{k} = \binom{n}{n-k}$$

Proof: The above is true since the number of ways to select k items out of n is the same as the number of ways to choose $n-k$ items to *leave out*.

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

Proof: To verify the above statement, consider whether the nth element in the total set is selected or not. If the nth element is selected, we can choose $k-1$ elements from the remaining $n-1$, which leads to $\binom{n-1}{k-1}$ distinct cases. If the nth is rejected, we can select k items from the remaining $n-1$, leading to $\binom{n-1}{k}$. The total is $\binom{n-1}{k} + \binom{n-1}{k-1}$, which is the number of ways to select k from n total items.

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + ... + \binom{n}{n} = 2^n.$$

Proof: Consider the number of subsets, ranging from 0 to n elements, we can form from the total set of n items. Separately, each item is either selected or not selected, so there are two possibilities for each item, producing a grand total of $\underbrace{2 * 2 * 2 * ... * 2}_{\text{n times}} = 2^n$ total ways to produce a subset. We can also add up the number of ways to select 0, 1, 2, and all the way up to n elements from the total set. This is equal to $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + ... + \binom{n}{n}$. Thus, the left hand side and right hand side of the equation are equal.

Combinations can be used to solve a wide range of problems in the AMC spectrum, and can be a stellar tool in conjunction with permutations.

Example 3

Fifteen distinct points are designated on $\triangle ABC$: the 3 vertices A, B, and C; 3 other points on side $\overline{AB}$; 4 other points on side $\overline{BC}$; and 5 other points on side $\overline{CA}$. Find the number of triangles with positive area whose vertices are among these 15 points. (AIME I 2017)

Solution: There are 15 total possible vertices and we need to select 3 of them to form a triangle, so there are $\binom{15}{3} = \frac{15!}{12!3!} = 455$ ways to accomplish this. However, the three selected points cannot all reside on the same line since they cannot form a triangle if they do. Note that there are $3 + 3 = 6$

points on $\overline{AB}$, $3+4=7$ points on $\overline{BC}$, and $3+5=8$ points on $\overline{CA}$.

The number of ways these three points can be on $\overline{AB}$ is $\binom{5}{3}=10$. The number of ways to select three points on $\overline{BC}$ is $\binom{6}{3}=20$. Finally, the number of ways to select three vertices on $\overline{CA}$ is $\binom{7}{3}=35$. Thus, we need to get rid of $10+20+35=65$ cases from the original count of 455, which leads to an answer of $455-65=\boxed{390}$.

Example 4

The nine delegates to the Economic Cooperation Conference include 2 officials from Mexico, 3 officials from Canada, and 4 officials from the United States. During the opening session, three of the delegates fall asleep. Assuming that the three sleepers were determined randomly, the probability that exactly two of the sleepers are from the same country is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. (AIME I 2015)

Solution: Note that the total number of ways to select 3 sleepers out of 9 delegates is $\binom{9}{3}=84$. We contemplate how many of these selections satisfies the property that exactly two of the sleepers are from the same country. We consider three cases.

- **Two delegates from US fall asleep**. The number of ways to select two sleepers from the US is $\binom{4}{2}=6$. Then, the remaining sleeper can be selected from the $9-4=5$ non-US delegates. This leads to a total of $6*5=30$ cases.
- **Two delegates from Canada fall asleep**. The number of ways to select two sleepers from Canada is $\binom{3}{2}=3$. Then, the remaining sleeper can be selected from $9-3=6$ non-Canada delegates. This leads to a total of $3*6=18$ cases.
- **Two delegates from Mexico fall asleep**. The number of ways to select two sleepers from Mexico is $\binom{2}{2}=1$. Then, the remaining sleeper can be selected from $9-2=7$ non-Mexico delegates. This leads to a total of $1*7=7$ cases.

The number of cases with two sleepers from the same country is $30+18+7=55$. Since the total number of ways to select sleepers is 84, the probability that two sleepers are from the same conutry is $\frac{55}{84}$. Thus, as $m=55$ and $n=84$, $m+n=\boxed{139}$.

Binomial Theorem

Binomial Theorem

Let a and b be complex or real numbers and $n>0$ be an integer. Then,

$$(a+b)^n=\binom{n}{0}a^n+\binom{n}{1}a^{n-1}b^1+...+\binom{n}{n-1}a^1b^{n-1}+\binom{n}{n}b^n.$$

Note that $\binom{n}{k}=\frac{n!}{k!(n-k)!}$.

We can crystallize the Binomial Theorem more succinctly as follows:

$$(a+b)^n=\sum_{i=0}^{n}\binom{n}{i}a^{n-i}b^i.$$

A mainstream proof of the Binomial Theorem revolves around the Principle of Mathematical Induction, which is widely available online. We will deploy the Binomial Theorem to solve a vibrant array of problems with applications from algebraic reduction to clever counting strategies.

Example 5

Prove that $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + ... + \binom{n}{n} = 2^n$.

Solution: Use the Binomial Theorem equation and set $a = 1$ and $b = 1$. Then,

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b^1 + ... + \binom{n}{n-1}a^1b^{n-1} + \binom{n}{n}b^n$$
$$(1+1)^n = \binom{n}{0}1^n + \binom{n}{1}1^{n-1}1^1 + ... + \binom{n}{n-1}1^11^{n-1} + \binom{n}{n}1^n$$
$$2^n = \binom{n}{0} + \binom{n}{1} + ... + \binom{n}{n-1} + \binom{n}{n}.$$

This culminates the proof.

Example 6

Find the sum of the roots, real and non-real, of the equation $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$, given that there are no multiple roots. (AIME 2001)

Solution: We note that the leading term of $(\frac{1}{2} - x)^{2001}$ is $\binom{2001}{2001}(-x)^{2001} = -x^{2001}$, which cancels out with x^{2001}. In the remaining expression, the term with the highest degree is x^{2000}. According to Vieta's Formuals, to find the sum of the roots, we need the coefficient of x^{2000} and the coefficient of x^{1999}, divide the two numbers, and take the negative of that.

The term with degree 2000 is equal to $\binom{2001}{1}(\frac{1}{2})^1(-x)^{2020} = \frac{2001x^{2020}}{2}$. The term with degree 1999 is equal to $\binom{2001}{2}(-x)^{1999}(\frac{1}{2})^2 = -2001 * 250x^{1999}$. The sum of the roots is the negative of the coefficient of x^{1999} divided by the coefficient of x^{2020}, or

$$-\frac{-2001 * 250}{\frac{2001}{2}} = 2001 * 250 * \frac{2}{2001} = \boxed{500}.$$

Example 7

What is the hundreds digit of 2011^{2011}? (AMC 10B 2011)

Solution: We note that we can write 2011^{2011} as $(2000 + 11)^{2011}$. Using Binomial Theorem, we can express this as

$$(2000+11)^{2011} = \binom{2011}{0}(2000)^{2011} + \binom{2011}{1}(2000)^{2010}(11)^1 + ... + \binom{2011}{2011}(11)^{2011}.$$

We note that all of the terms except for the last one are divisible by 1000. Thus, the last term will determine the hundreds digit. The last term is $11^{2011} = (1+10)^{2011}$. Using another application of the Binomial Theorem, we find that:

$$11^{2011} = (1+10)^{2011} \quad = \quad \binom{2011}{0} * 1^{2011} + \binom{2011}{1} * 1^{2010}10^1 + \binom{2011}{2}1^{2009}10^2 + ...$$

The rest of the terms are divisible by 1000, so we just have to consider the three terms written above to discover the hundreds digit. Note that the $\pmod{1000}$ terminology below represents modular

congruence, meaning that the two numbers on both sides of the $\equiv$ symbol have the same remainder when divided by 1000. Thus,

$$\begin{aligned}
11^{2011} &\equiv \binom{2011}{0} * 1^{2011} + \binom{2011}{1} * 1^{2010}10^1 + \binom{2011}{2} 1^{2009}10^2 \pmod{1000} \\
&\equiv 1 + 2011(10) + \frac{2011 * 2010}{2} * 100 \pmod{1000} \\
&\equiv 1 + 110 + (11 * 10) * 50 \pmod{1000} \\
&\equiv 111 + 5500 \pmod{1000} \\
&\equiv 611 \pmod{1000}.
\end{aligned}$$

Thus, the hundreds digit of this expression is $\boxed{6}$.

Exercises

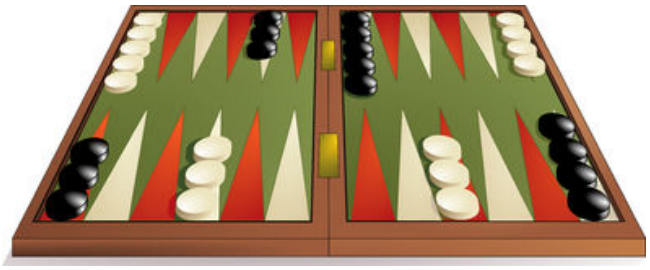
1. Compute the unit's digit of 9^{200}.
2. What is the units digit of $19^{19} + 99^{99}$? (AMC 8 2000)
3. A bubble tea shop sells 7 different flavors of tea. Three friends walk to the shop and each purchase a different flavor of tea. How many distinct orders can the 3 friends create?
4. Alex, Bryce, Cynthia, Daisy, and Ellie are in a line. How many ways can they be arranged in the line so that Cynthia and Daisy are standing right next to each other?
5. How many ways can five students be seated around a circular dinner table? Two seating arrangements are not distinct if one can be obtained by rotating the other.
6. In a game of *Among Us* with 7 players, 2 of the players are selected as imposters. How many possible pairs of imposters can be selected, if the order of selection does not matter?
7. A game uses a deck of n different cards, where n is an integer and $n \geq 6$. The number of possible sets of 6 cards that can be drawn from the deck is 6 times the number of possible sets of 3 cards that can be drawn. Find n. (AIME II 2005)
8. A student council must select a two-person welcoming committee and a three-person planning committee from among its members. There are exactly 10 ways to select a two-person team for the welcoming committee. It is possible for students to serve on both committees. In how many different ways can a three-person planning committee be selected? (AMC 10A 2013)
9. A dessert chef prepares the dessert for every day of a week starting with Sunday. The dessert each day is either cake, pie, ice cream, or pudding. The same dessert may not be served two days in a row. There must be cake on Friday because of a birthday. How many different dessert menus for the week are possible? (AMC 10B 2012)
10. Ten women sit in 10 seats in a line. All of the 10 get up and then reseat themselves using all 10 seats, each sitting in the seat she was in before or a seat next to the one she occupied before. In how many ways can the women be reseated? (AMC 12B 2009)
11. Two subsets of the set $S = \{a, b, c, d, e\}$ are to be chosen so that their union is S and their intersection contains exactly two elements. In how many ways can this be done, assuming that the order in which the subsets are chosen does not matter? (AMC 10A 2008)
12. At a gathering of 30 people, there are 20 people who all know each other and 10 people who know no one. People who know each other hug, and people who do not know each other shake hands. How many handshakes occur within the group? (AMC 10A 2017)

13. Five friends sat in a movie theater in a row containing 5 seats, numbered 1 to 5 from left to right. (The directions "left" and "right" are from the point of view of the people as they sit in the seats.) During the movie Ada went to the lobby to get some popcorn. When she returned, she found that Bea had moved two seats to the right, Ceci had moved one seat to the left, and Dee and Edie had switched seats, leaving an end seat for Ada. In which seat had Ada been sitting before she got up? (AMC 10A 2016)

14. How many subsets of $\{2, 3, 4, 5, 6, 7, 8, 9\}$ contain at least one prime number? (AMC 10B 2018)

15. Seven distinct pieces of candy are to be distributed among three bags. The red bag and the blue bag must each receive at least one piece of candy; the white bag may remain empty. How many arrangements are possible? (AMC 10B 2010)

16. Central High School is competing against Northern High School in a backgammon match. Each school has three players, and the contest rules require that each player play two games against each of the other school's players. The match takes place in six rounds, with three games played simultaneously in each round. In how many different ways can the match be scheduled? (AMC 10A 2013)



17. Ten Avengers are plotting a heist where they must retrieve infinity stones from three locations, with one group assigned to each location. They split into three groups, with 4 Avengers heading to New York, 2 heading to Asgard, and 4 heading to Morag. Suppose that two of the Avengers, Captain America and Tony Stark, must be in the same group, while another two of the Avengers, Thor and Tony Stark, insist on being in different groups. How many distinct ways are there to assign the Avengers to locations?

18. Eight points are chosen on a circle, and chords are drawn connecting every pair of points. No three chords intersect in a single point inside the circle. How many triangles with all three vertices in the interior of the circle are created? (AMC 10A 2010)

19. Eight people are sitting around a circular table, each holding a fair coin. All eight people flip their coins and those who flip heads stand while those who flip tails remain seated. What is the probability that no two adjacent people will stand? (AMC 10A 2015)

20. Three tiles are marked X and two other tiles are marked O. The five tiles are randomly arranged in a row. What is the probability that the arrangement reads $XOXOX$? (AMC 10A 2005)

21. How many ways can a merchant travel from $(0, 0)$ to $(6, 6)$ moving only one unit to the right or one unit upward each step? Note that any such path can be written as an arrangement of moves in the right direction R and moves in the up direction U, such as $RRRRRRUUUUUU$.

22. Allison has a coin which comes up heads $\frac{2}{3}$ of the time. She flips it 5 times. What is the probability that she sees more heads than tails? (HMMT 2010)

23. Newton and Leibniz are playing a game with a coin that comes up heads with probability p. They take turns flipping the coin until one of them wins with Newton going first. Newton wins

if he flips a heads and Leibniz wins if he flips a tails. Given that Newton and Leibniz each win the game half of the time, what is the probability p? (HMMT 2010)

24. A poll shows that 70% of all voters approve of the mayor's work. On three separate occasions a pollster selects a voter at random. What is the probability that on exactly one of these three occasions the voter approves of the mayor's work? (AMC 10B 2008)

25. Three married couples on a cruise together are perusing a pamphlet that describer three different shore excursions. Collectively, they want to "try everything", so at least one person goes on each excursion. In how many ways can this be done if no one wants to be on the same excursion with his/her spouse? (MN Math League 13/14)

26. The expression
$$(x+y+z)^{2006}+(x-y-z)^{2006}$$
is simplified by expanding it and combining like terms. How many terms are in the simplified expression? (AMC 12A 2006)

27. For nonnegative integers a and b with $a+b \leq 6$, let $T(a,b) = \binom{6}{a}\binom{6}{b}\binom{6}{a+b}$. Let S denote the sum of all $T(a,b)$, where a and b are nonnegative integers with $a+b \leq 6$. Find the remainder when S is divided by 1000. (AIME I 2017)

28. Compute the number of permutations $x_1, ..., x_6$ of the integers $1, ..., 6$ such that $x_{i+1} \leq 2x_i$ for all i, $1 \leq 6$. (ARML 2016)

29. Bobby, Peter, Greg, Cindy, Jan, and Marcia line up for ice cream. In an *acceptable lineup*, Greg is ahead of Peter, Peter is ahead of Bobby, Marcia is ahead of Jan, and Jan is ahead of Cindy. For example, the lineup with Greg in front, followed by Peter, Marcia, Jan, Cindy, and Bobby, in that order, is an acceptable lineup. Compute the number of acceptable lineups. (ARML 2014)

30. How many 15-letter arrangements of 5 A's, 5 B's, and 5 C's have no A's in the first 5 letters, no B's in the next 5 letters, and no C's in the last 5 letters? (AMC 12A 2003)